

4(a)(i)	RNA polymerase ;	1
4(a)(ii)	phosphodiester ;	1
4(b)(i)	5 ;	1
4(b)(ii)	any two from: 1 other strand is not the <u>template</u> strand / only the <u>template</u> strand is transcribed ; 2 (if transcribed) sequence of, nucleotides / bases, in (m)RNA codes for a different sequence of amino acids ; A sequence of codons A primary transcript for RNA 3 (if transcribed) results in a non-functional polypeptide / AW ; A codes for a polypeptide with different function 4 translation would be a waste of energy / translation will not occur (as no signalling sequence) ; 5 any suggestion of a possible consequence to structure of collagen ; 6 (m)RNA is single stranded ; 7 AVP ; e.g. RNA polymerase synthesises RNA in the 5' to 3' direction	2
4(c)(i)	ref. to (mitochondrial) envelope / two membranes / double membrane / inner membrane and outer membrane ; R cell or cell surface membrane inner membrane is folded / crista(e) ; (70S) ribosomes ; R 80S ribosomes (mitochondrial) matrix ; I DNA	2
4(c)(ii)	0.6 (μm) ; R 0.60 (only one significant figure required)	1

Question	Answer	Marks
4(d)	any three from: 1 active transport, uses ATP / moves (ions) against concentration gradient ; 2 facilitated diffusion, passive / no ATP / no energy / with the gradient / down concentration gradient ; 3 protein, carrier / pump, changes shape / <i>ref. to</i> conformation change / has binding site(s) ; 4 channel protein / pore protein, for facilitated diffusion (only) ; I carrier protein 5 AVP ; e.g. <i>ref. to</i> ionophore(s) / hydrophilic pore in channel proteins / specificity of carrier <i>or</i> channel protein(s) <i>if mp1 and mp2 are not awarded allow one mark for naming active transport <u>and</u> facilitated diffusion</i>	3

Question	Answer	Marks
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